Digital Electronics & Systems

Lecture 1
Introduction

23ECE205

Digital Electronics & Systems

- **TEXT BOOK:** Stephen Brown, Zvonko Vranesic, "Fundamentals of Digital logic with Verilog Design", Tata McGraw Hill Publishing Company Limited, Special Indian Edition, 2007.

Course Objectives

- To understand the fundamentals of Boolean Logic and the building blocks of digital circuits
- To introduce the abstraction of simple practical problems into Boolean Logic and their efficient implementation and to introduce the fundamentals of design with combinational and sequential subsystems

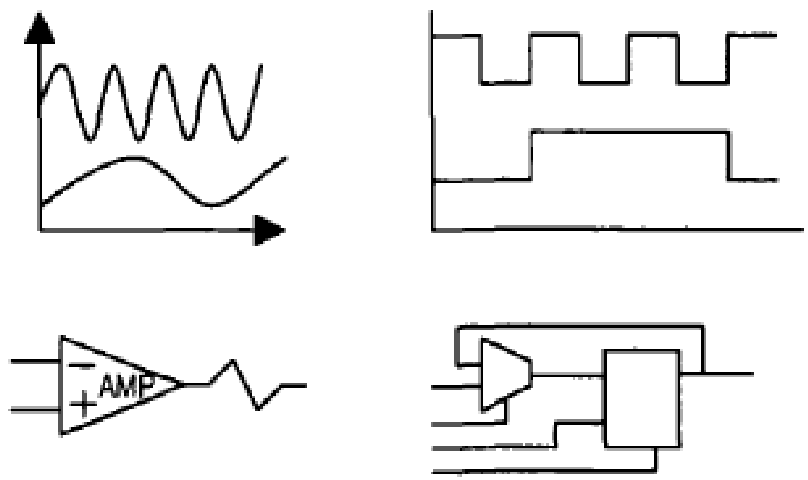
Course Outcomes

- **CO1:** Able to frame Boolean equations for solving a simple real-life problems and realize them using gate-level building blocks
- **CO2:** Able to apply minimization techniques for efficient Boolean logic implementation
- **CO3:** Able to realize digital blocks using combinational and sequential subsystems
- **CO4:** Able to design using state machine descriptions for practical real-life engineering problems

Evaluation Scheme

ANALOG VS DIGITAL

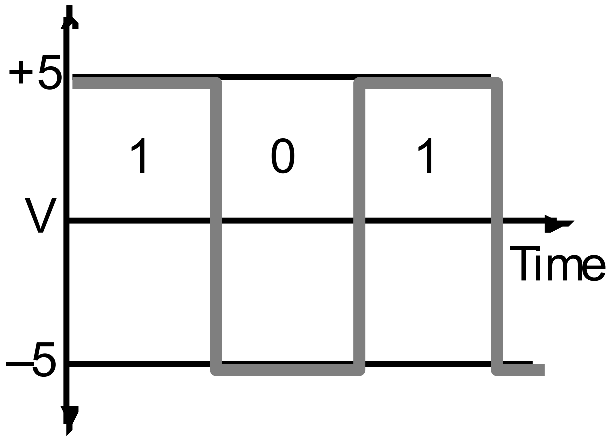
- Analog devices and systems: Process analog signals (time-varying signals that can take any value across a continuous range known as dynamic range)
- Digital devices and systems: Process digital signals (analog signals that are modeled as having at any time one of two discrete values)

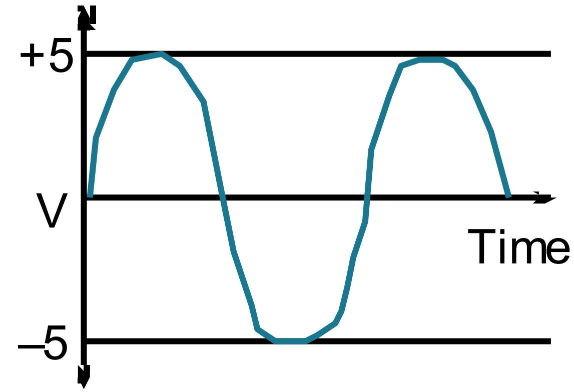


a. Analog Circuit. b. Digital Circuit.

Digital Systems

Digital vs. Analog Waveforms



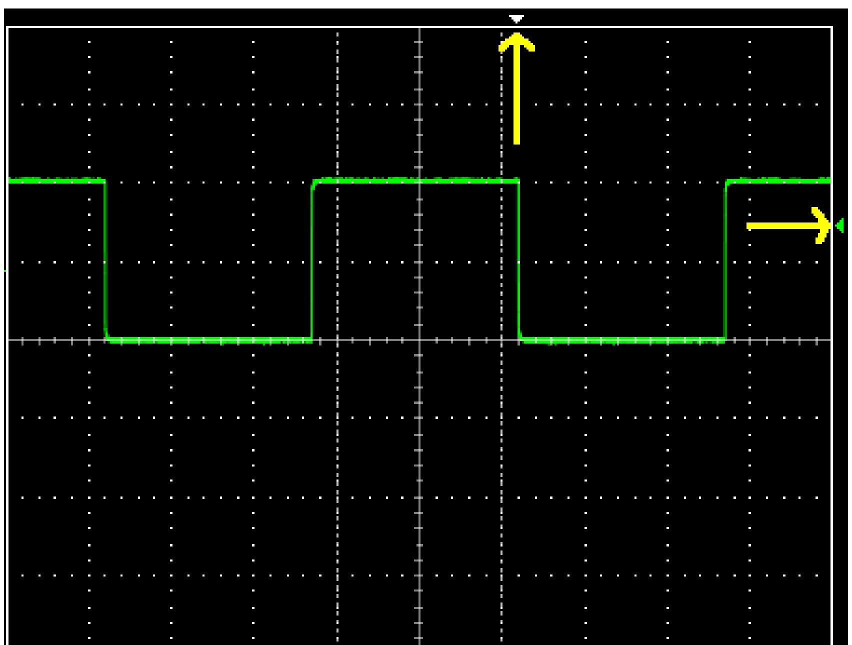


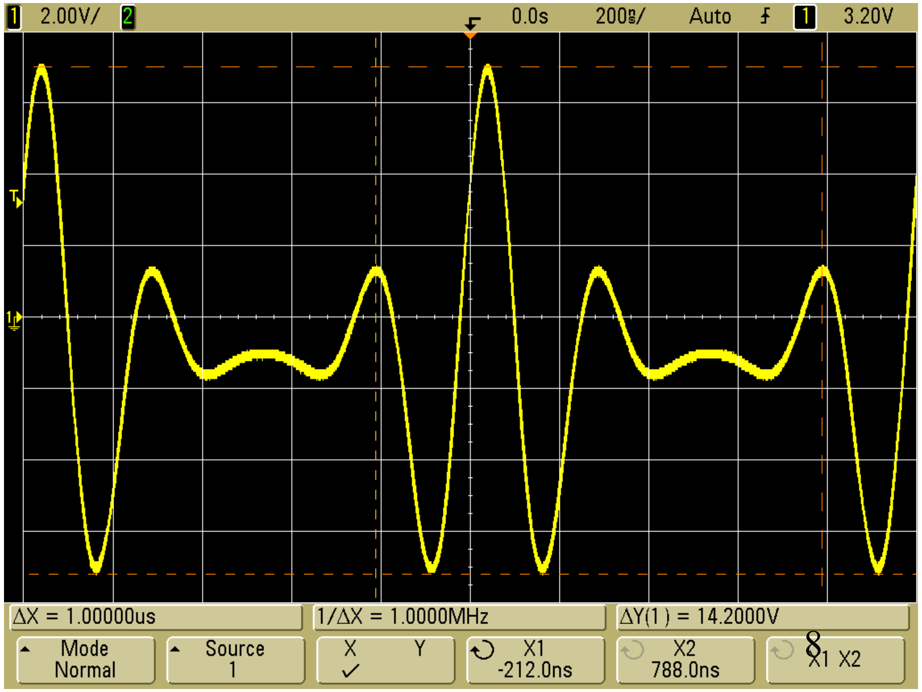
Digital:
only assumes discrete values

Analog:
values vary over a broad range continuously

Digital Systems cntd..

- A digital **signal** is a signal that can only have discrete values in time
  - Most common are **binary** digital signals, where only two values are allowed often designated as 0 and 1
- The opposite is analog signals that can take infinite values





Boolean Algebra and Logical Operators

Algebra: **variables, values, operations**

In Boolean algebra, the values are the symbols 0 and 1
If a logic statement is false, it has value 0
If a logic statement is true, it has value 1

Operations: AND, OR, NOT

X	Y	X AND Y
0	0	0
0	1	0
1	0	0
1	1	1

X	Y	X OR Y
0	0	0
0	1	1
1	0	1
1	1	1

X	NOT X
0	1
1	0

Variables & Functions

- Logic circuits form the foundation of digital systems
- Binary logic circuits perform operations of binary signals
- Let's consider the simplest element logic circuit: switch.
  - A switch can be in two states: "open" and "close".
  - Graphical symbol for a switch

$x = 0$

State of the switch can be given as a binary variable x

Representations of Digital Design: Switches

Normally Open

A switch connects two points under control signal.

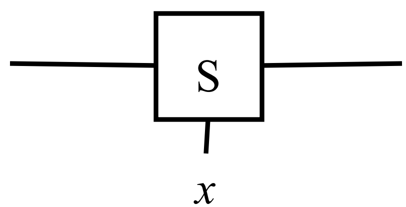
- when the control signal is 0 (false), the switch is open
- when it is 1 (true), the switch is closed

We will use Normally open switch in our discussion

A switch connects two points under control signal.

- when the control signal x is 0 (false), the switch is open
- when it is 1 (true), the switch is closed

(a) Two states of a switch

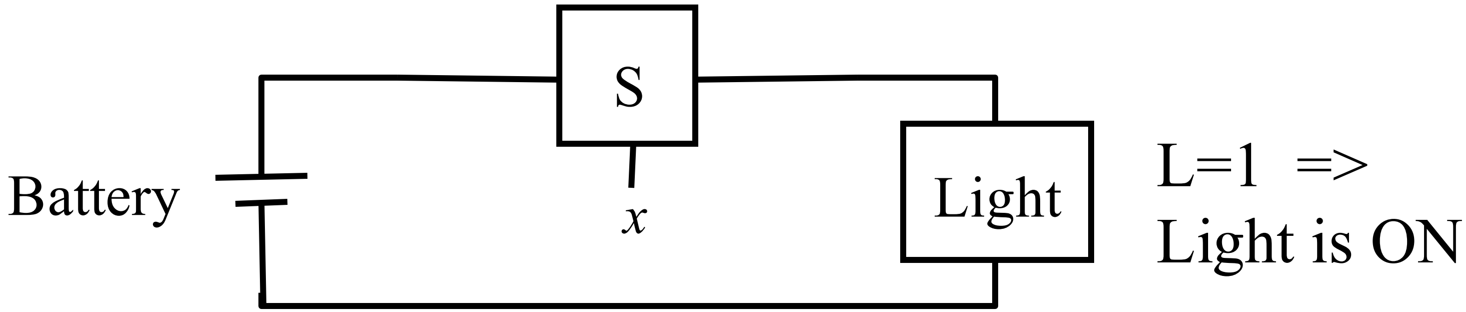


(b) Symbol for a switch

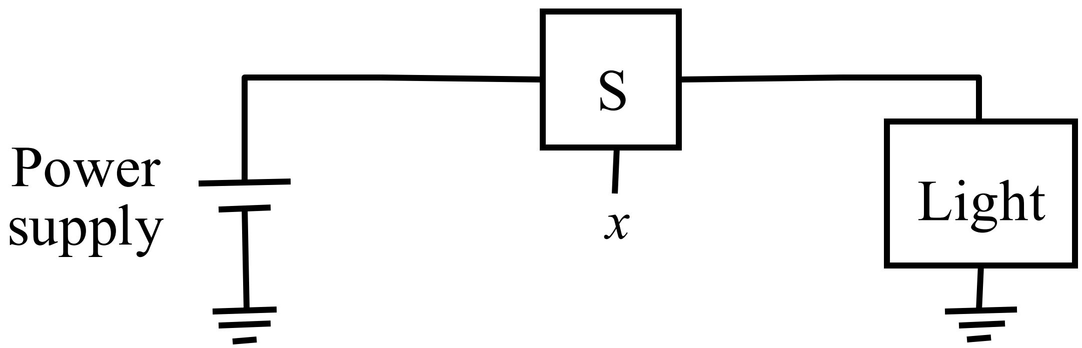
A simple application using a switch : "lightbulb"
State of the switch can be given as a binary variable x s.t.:

- $x = 0$ when switch is open, and
- $x = 1$ when switch is closed.

Figure: A binary switch.



(a) Simple connection to a battery



(b) Using a ground connection as the return path

Figure : A light controlled by a switch.

Logic expression/Logic Function

State of Light is function of input variable x.

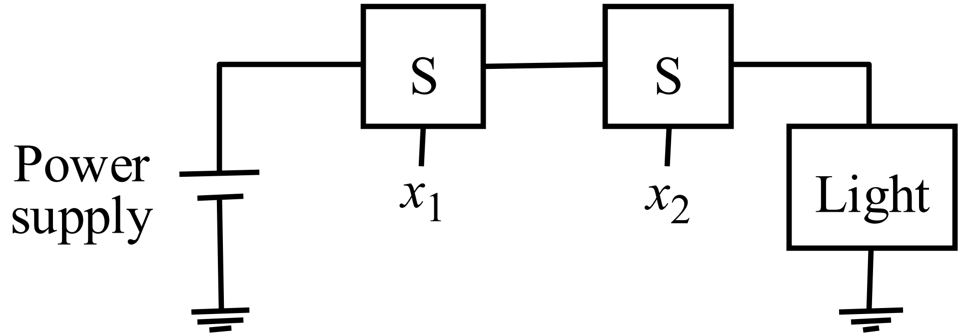
- L=1 if x=1
- L=0 if x=0

L(x)=x

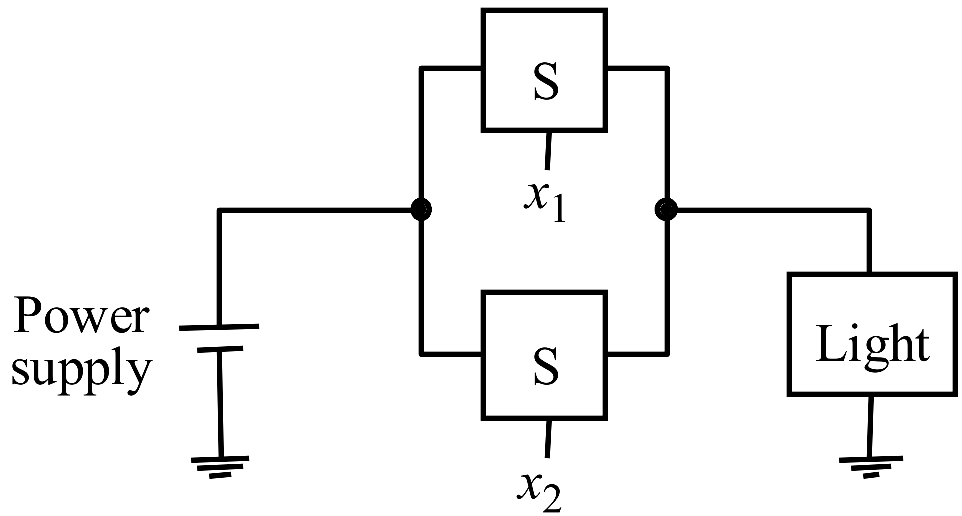
L(x)=x is a logic expression/function

Here only one switch was controlling Light.

If there are two switches then?



(a) The logical AND function (series connection)



(b) The logical OR function (parallel connection)

Figure : Two basic functions.

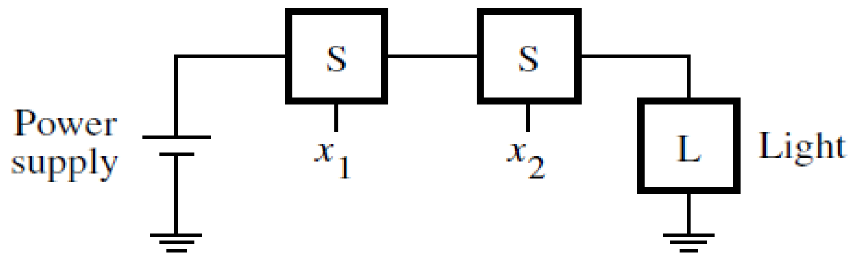
Series Connection => logical AND operation

$$L(x_1, x_2) = x_1 \cdot x_2$$

$$\text{where} \quad L = 1 \text{ if } x_1 = 1 \text{ and } x_2 = 1,$$
$$L = 0 \text{ otherwise.}$$

The symbol

- Is called **AND** operator



(a) The logical AND function (series connection)

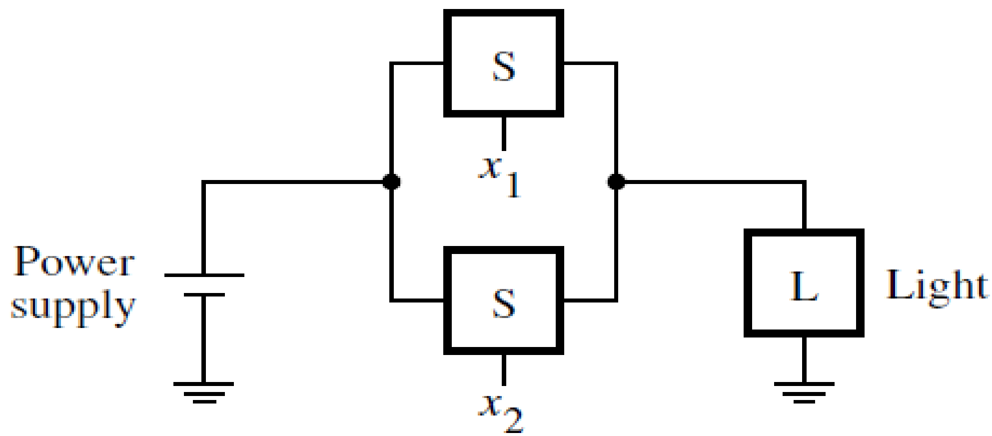
Parallel Connection => logical OR operation

$$L(x_1, x_2) = x_1 + x_2$$

where $L = 1$ if $x_1 = 1$ or $x_2 = 1$ or if $x_1 = x_2 = 1$,

$L = 0$ if $x_1 = x_2 = 0$.

Symbol $+$ is called **OR** operator



(b) The logical OR function (parallel connection)

Here three switches are controlling the light in more complex way. Can you identify the relation?

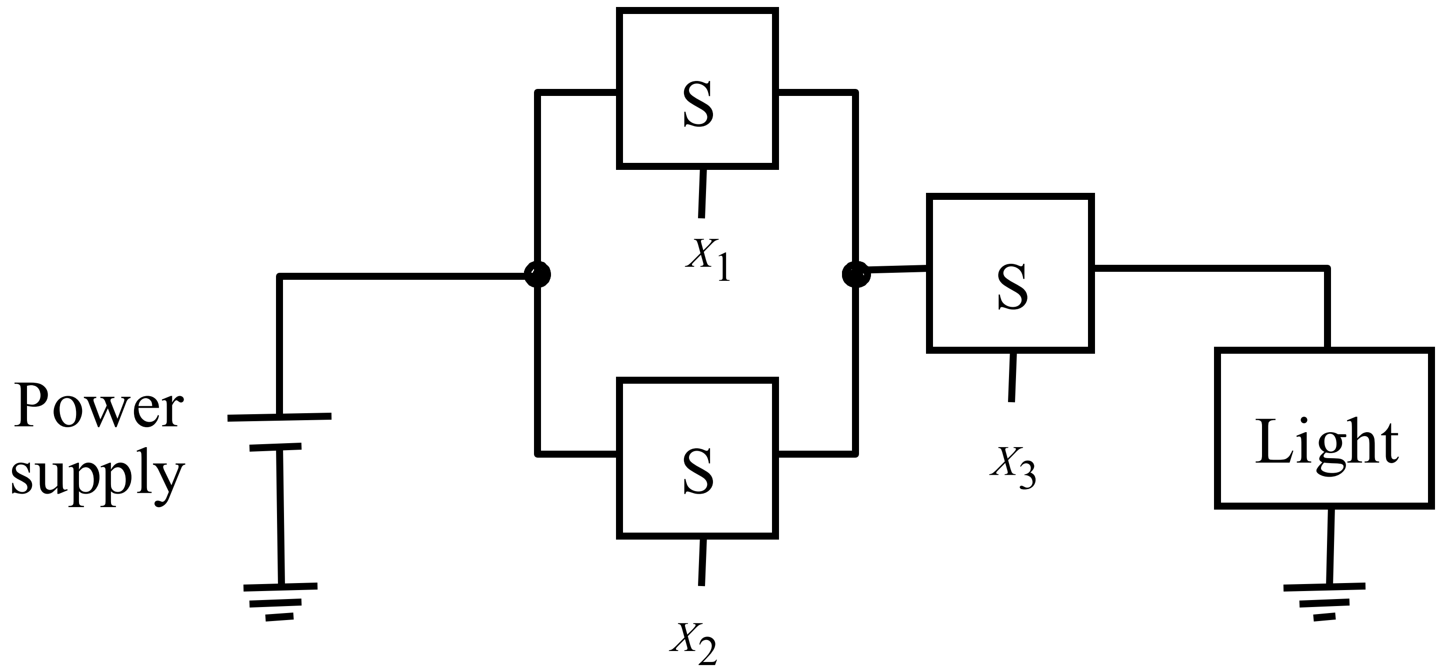
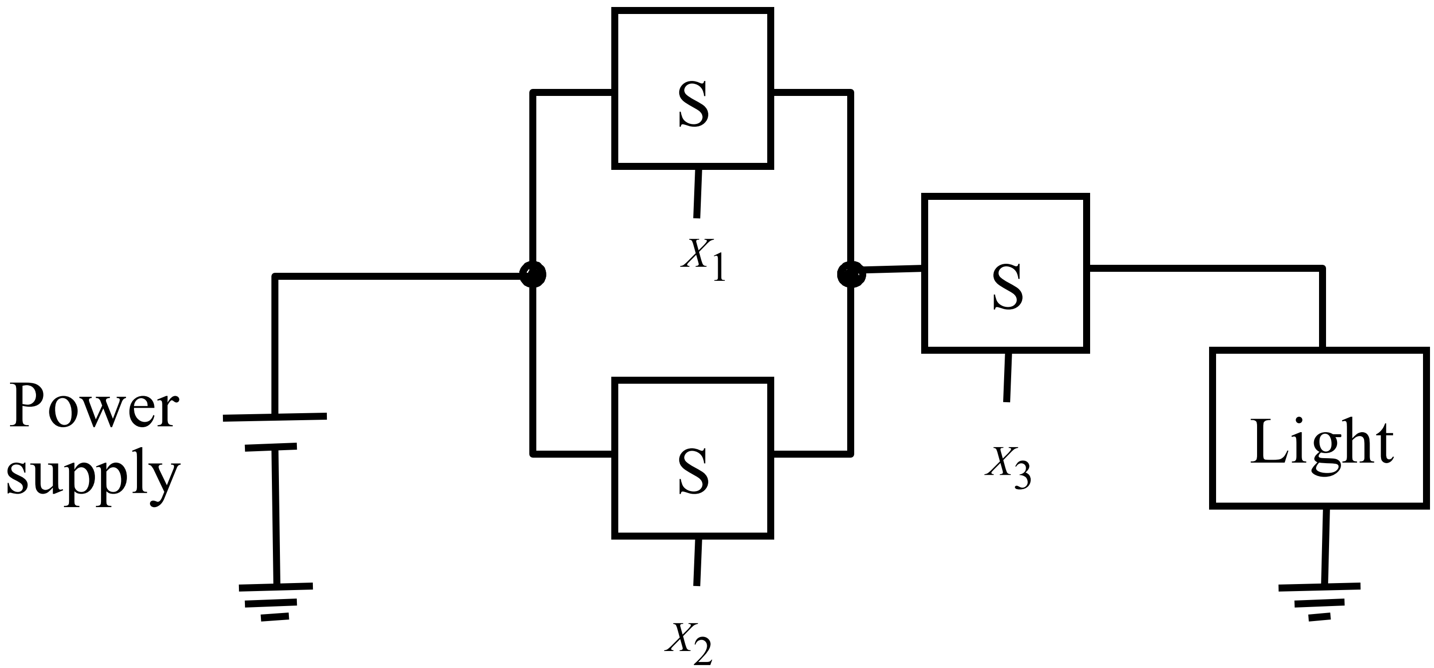


Figure: A series-parallel connection.

Here the series parallel connection of switches realizes the logic function

$$L(x_1, x_2, x_3) = (x_1 + x_2) \cdot x_3$$



The light is on if $x_3 = 1$ and, at the same time, at least one of the x_1 or x_2 inputs is equal to 1.

Figure : A series-parallel connection.

Try implementation using switches

- (A+B).(C+D)
- A.(B+C).D

LECTURE 2

- Till now we saw some positive action takes place when a switch is closed, such as turning the light on.
- It is equally interesting and useful to consider the possibility that a positive action takes place when a switch is opened

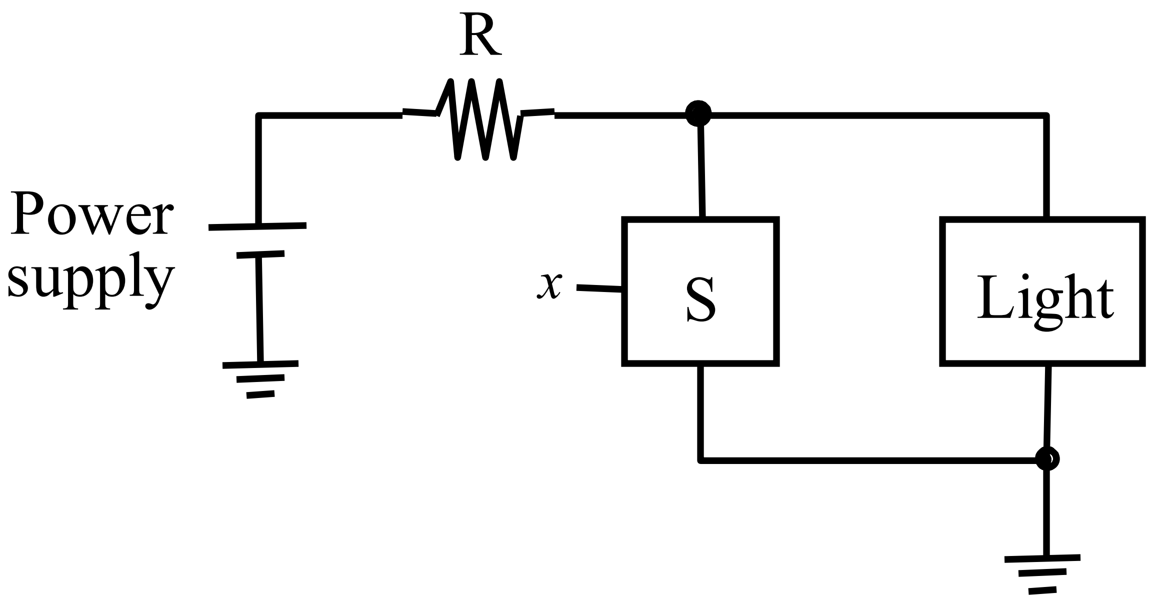


Figure : An inverting circuit.

Complement Operation

$$L(x) = \overline{x}$$

where $\quad L = 1 \text{ if } x = 0,$

$\quad L = 0 \text{ if } x = 1$

Symbol $\overline{}$ over a variable called **NOT** operator.

$$\overline{x} = x' = !x = \sim x$$

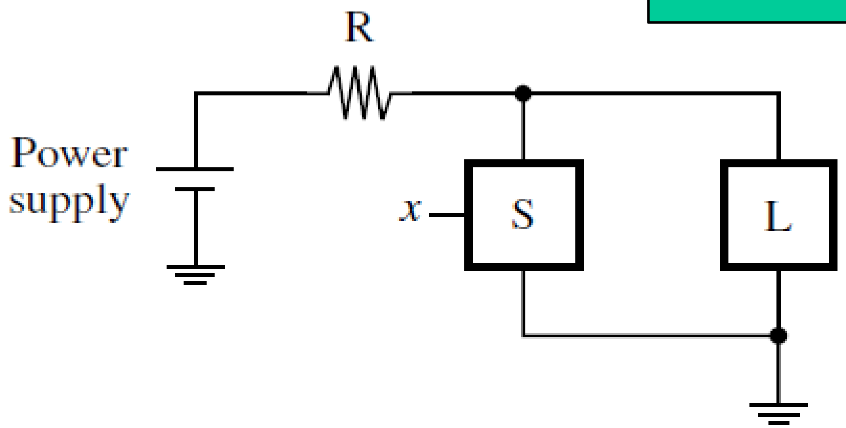


Figure 2.5 An inverting circuit.

Try implementation using switches

- ((A.B)+C).(D)
- (A+B)’

Truth Tables

Useful aid for giving information for a logic function.

x_1	x_2	$x_1 \cdot x_2$	$x_1 + x_2$
0	0	0	0
0	1	0	1
1	0	0	1
1	1	1	1
		AND	OR

Figure: A truth table for the AND and OR operations.

Truth Table for 3 input variables

x_1	x_2	x_3	$x_1 \cdot x_2 \cdot x_3$	$x_1 + x_2 + x_3$
0	0	0	0	0
0	0	1	0	1
0	1	0	0	1
0	1	1	0	1
1	0	0	0	1
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Figure : Three-input AND and OR operations.

Logic Gates

- Three basic logic operations can be used to implement logic functions of any complexity.
- Each logic operation can be implemented using Logic Gate.
- A logic gate has one or more inputs and one output that is a function of its inputs.
- It is often convenient to describe a logic circuit by drawing a circuit diagram, or *schematic*, consisting of graphical symbols representing the logic gates.

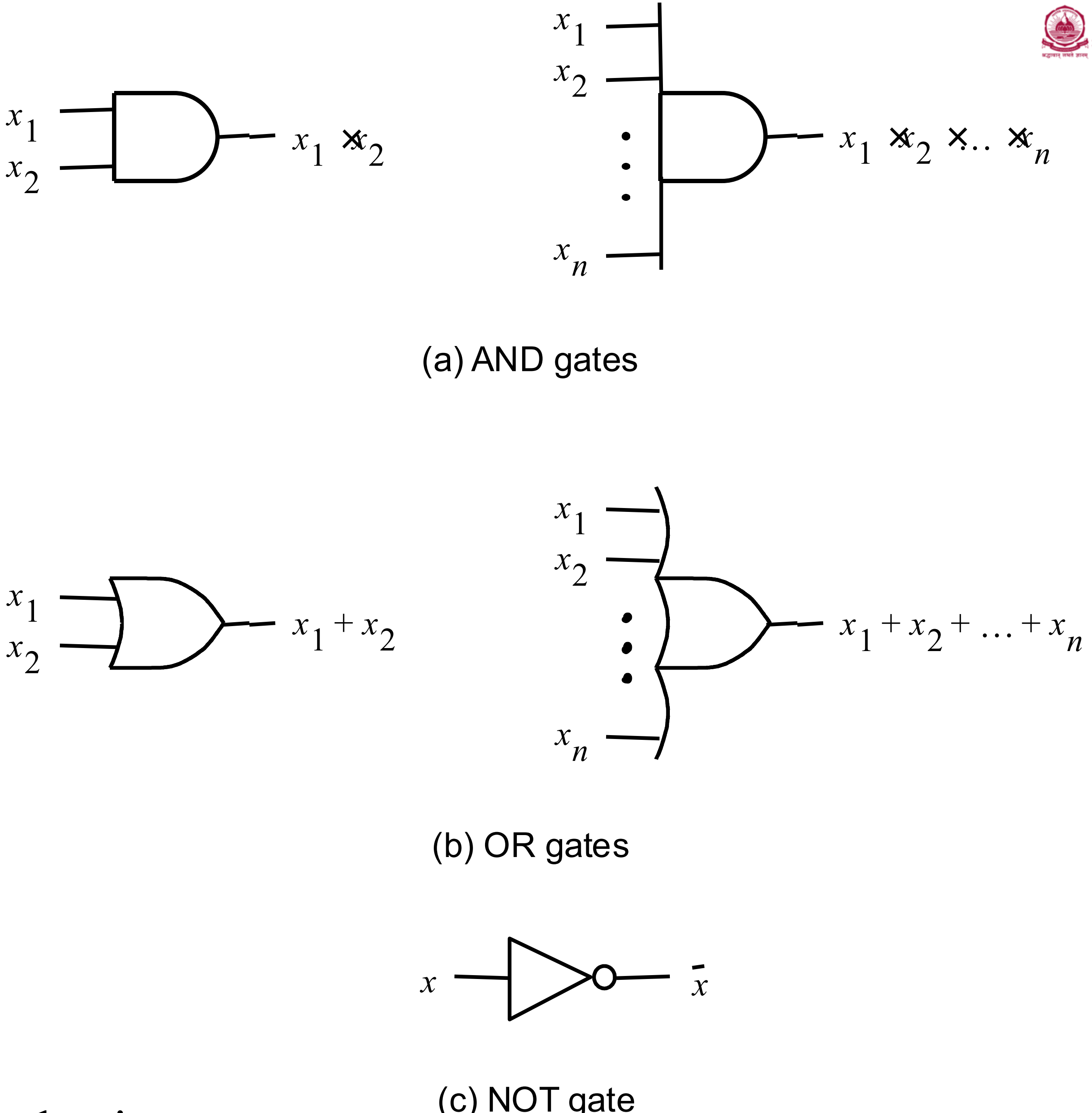
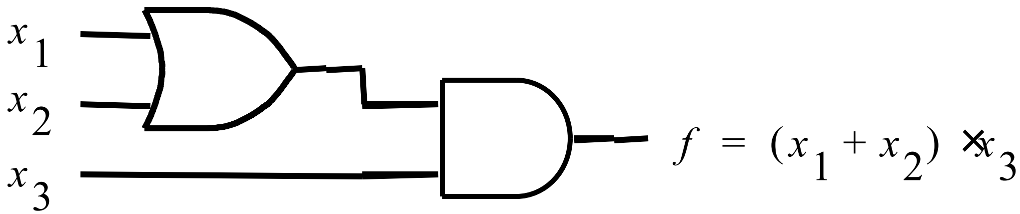


Figure: The basic gates.

A Larger circuit is implemented by Network of Gates



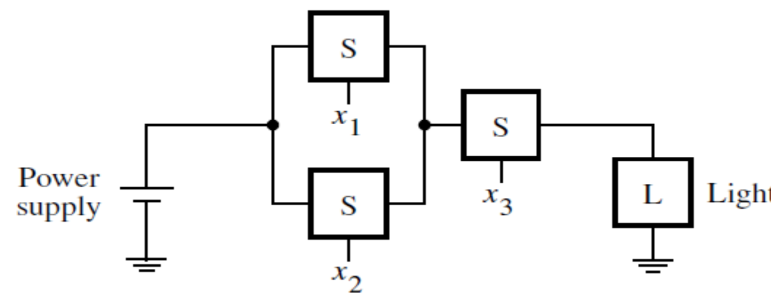
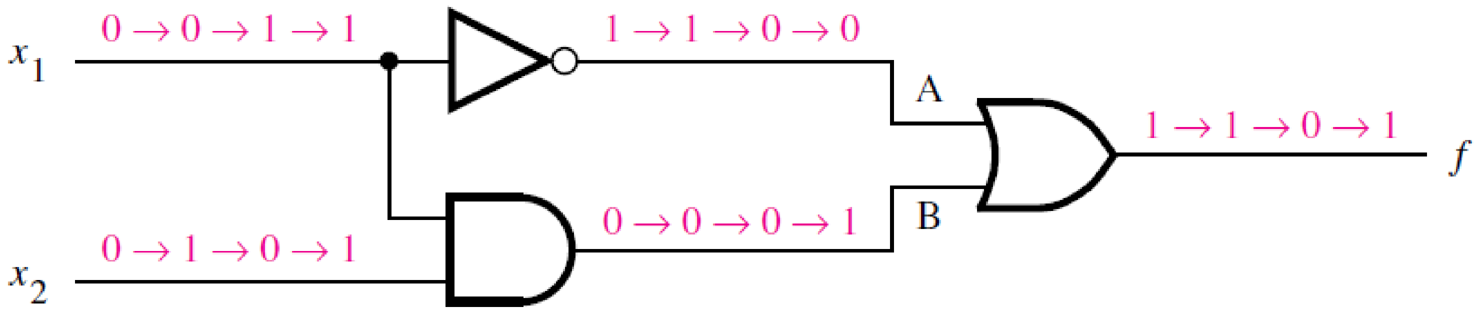


Figure: The function from Figure of switches .

Analysis vs Synthesis of Logic Networks

- Analysis Process: For an existing logic network, it must be possible to determine the function performed by the network. This task is referred to as the *analysis* process.
- Synthesis Process: The reverse task of designing a new network that implements a desired functional behavior is referred to as the *synthesis* process.
- The analysis process is straightforward and much simpler than the synthesis process

Analysis example



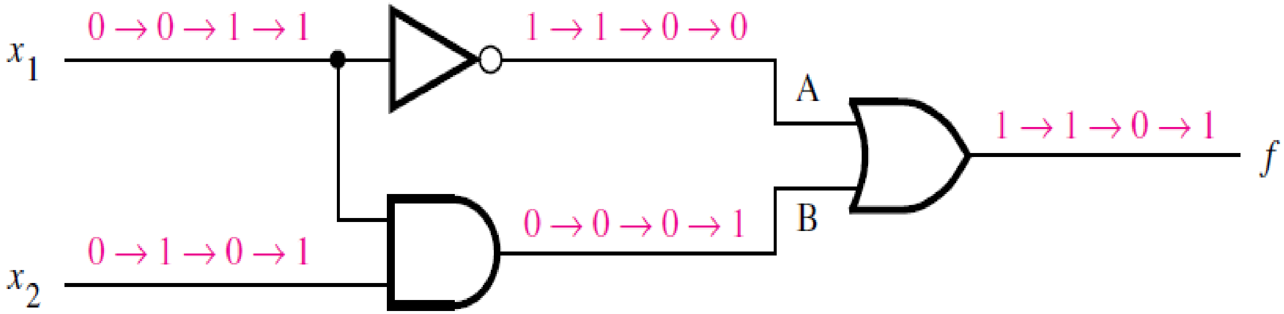
(a) Network that implements $f = \bar{x}_1 + x_1 \cdot x_2$

x_1	x_2	$f(x_1, x_2)$
0	0	1
0	1	1
1	0	0
1	1	1

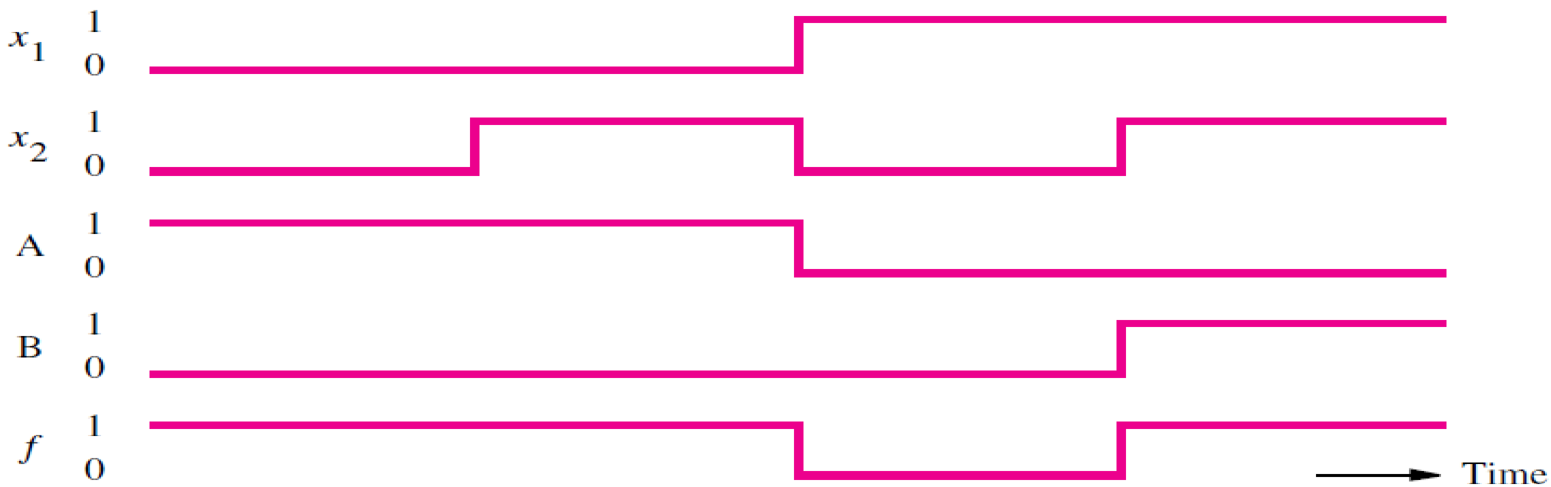
(b) Truth table for f

Check how signal A &B Are changing with input x1 & x 2

Timing Diagram: Functional Behavior



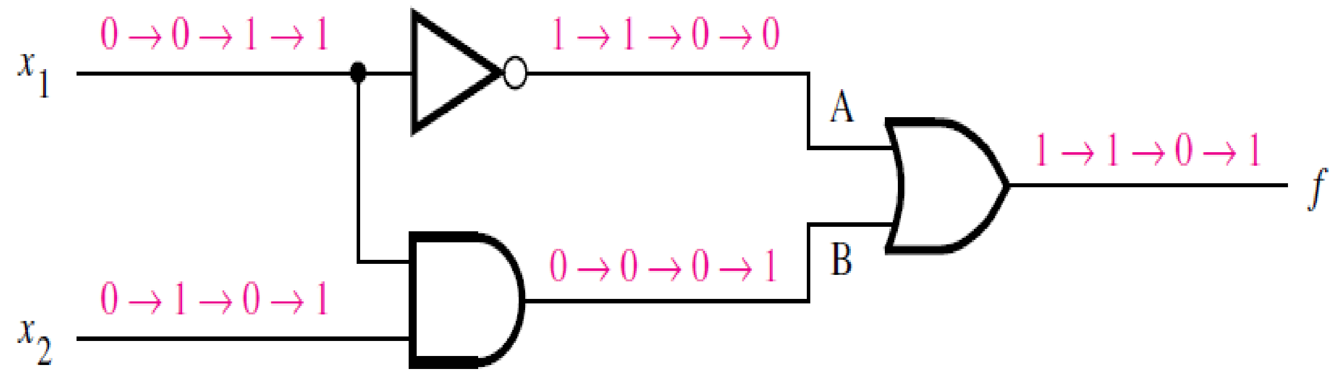
(a) Network that implements $f = \bar{x}_1 + x_1 \cdot x_2$



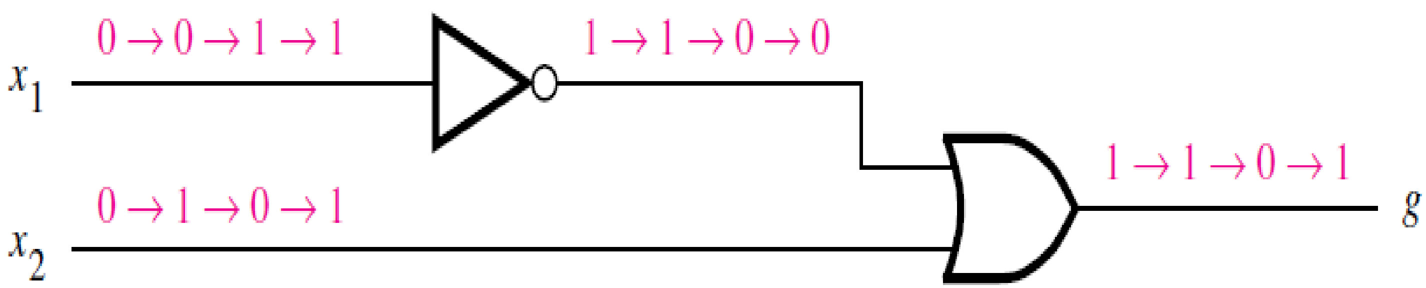
(c) Timing diagram

Check how signal A &B
Are changing with input
x1 & x 2

Functionally Equivalent Networks



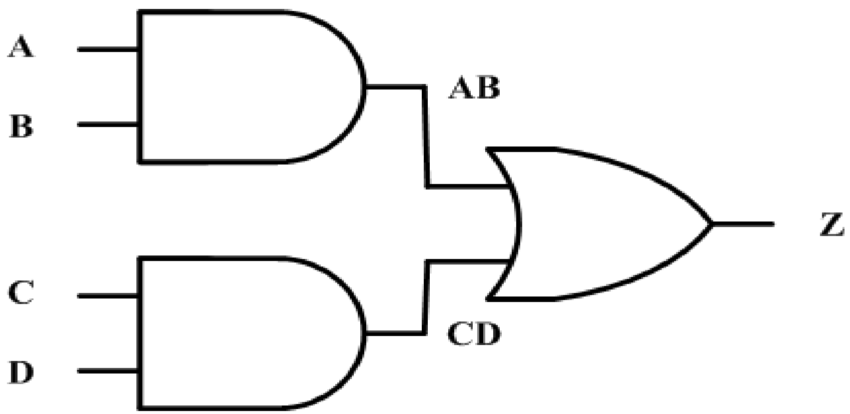
(a) Network that implements $f = \bar{x}_1 + x_1 \cdot x_2$



(d) Network that implements $g = \bar{x}_1 + x_2$

Truth table with 4 input variables

- we can also construct truth tables for any function.



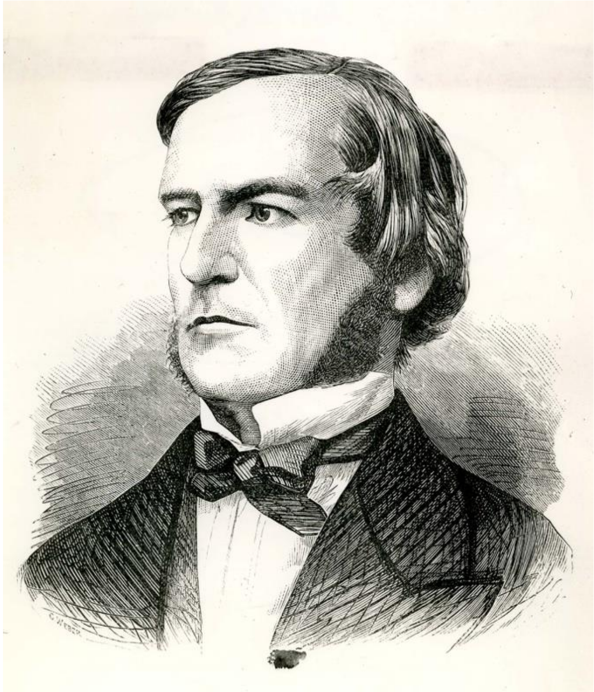
A	B	C	D	Z	AB	CD
0	0	0	0	0	0	0
0	0	0	1	0	0	0
0	0	1	0	0	0	0
0	0	1	1	1	0	1
0	1	0	0	0	0	0
0	1	0	1	0	0	0
0	1	1	0	0	0	0
0	1	1	1	1	0	1
1	0	0	0	0	0	0
1	0	0	1	0	0	0
1	0	1	0	0	0	0
1	0	1	1	1	0	1
1	1	0	0	1	1	0
1	1	0	1	1	1	0
1	1	1	0	1	1	0
1	1	1	1	1	1	1

Boolean Algebra

- Boolean Algebra
  - Axioms of Boolean Algebra
  - Single Variable Theorem
  - Two and Three variable Properties
  - Complements and Duals

Boolean Algebra

- In 1849 George Boole published a scheme for the algebraic description of processes involved in logical thought and reasoning . Subsequently, this scheme and its further refinements became known as *Boolean algebra*.
- It was almost 100 years later that this algebra found application in the engineering sense. In the late 1930s Claude Shannon showed that Boolean algebra provides an effective means of describing circuits built with switches .The algebra can be used to describe logic circuits.



Boolean Algebra:Axioms

1a $$0 \cdot 0 = 0 \quad (1)$$
1b $$1 + 1 = 1 \quad (2)$$

2a $$1 \cdot 1 = 1 \quad (3)$$
2b $$0 + 0 = 0 \quad (4)$$

3a $$0 \cdot 1 = 1 \cdot 0 = 0 \quad (5)$$
3b $$0 + 1 = 1 + 0 = 1 \quad (6)$$

4a $$\text{If } x = 0 \quad \text{then} \quad \overline{x} = 1 \quad (7)$$
4b $$\text{If } x = 1 \quad \text{then} \quad \overline{x} = 0 \quad (8)$$

Boolean Algebra: Single Variable Theorems

5a $$x \cdot 0 = 0 \quad (9)$$
5b $$x + 1 = 1 \quad (10)$$

6a $$x \cdot 1 = x \quad (11)$$
6b $$x + 0 = x \quad (12)$$

7a $$x \cdot x = x \quad (13)$$
7b $$x + x = x \quad (14)$$

8a $$x \cdot \overline{x} = 0 \quad (15)$$
8b $$x + \cdot\overline{x} = 1 \quad (16)$$

9 $$\overline{\overline{x}} = x \quad (17)$$

Principle of Duality

- Given a logic expression, its ***dual*** is obtained by replacing all + operators with · operators, and vice versa, and by replacing all 0s with 1s, and vice versa.
- The dual of any true statement (axiom or theorem) in Boolean algebra is also a true statement.

$$\mathbf{X + 0 = X}$$

$$\mathbf{X \cdot 1 = X}$$

OR +

A	0	RESULT
0	0	0
1	0	1

AND •

A	1	RESULT
0	1	0
1	1	1

LECTURE 3

Boolean Algebra: Two & Three variable Properties

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Commutativity:

10a $$x \cdot y = y \cdot x \tag{18}$$

10b $$x + y = y + x \tag{19}$$

Associativity:

11a $$x \cdot (y \cdot z) = (x \cdot y) \cdot z \tag{20}$$

11b $$x + (y + z) = (x + y) + z \tag{21}$$

Distributivity:

12a $$x \cdot (y + z) = x \cdot y + x \cdot z \tag{22}$$

12b $$x + y \cdot z = (x + y) \cdot (x + z) \tag{23}$$

Absorption:

13a $$x + x \cdot y = x \tag{24}$$

13b $$x \cdot (x + y) = x \tag{25}$$

Boolean Algebra: Two & Three variable Properties

Combining:

14a
14b

$$x \cdot y + x \cdot \overline{y} = x \quad (26)$$

$$(x + y) \cdot (x + \overline{y}) = x \quad (27)$$

Consensus:

17a
$$x \cdot y + y \cdot z + \overline{x} \cdot z = x \cdot y + \overline{x} \cdot z \quad (28)$$

17b
$$(x + y) \cdot (y + z) \cdot (\overline{x} + z) = (x + y) \cdot (\overline{x} + z) \quad (29)$$

Boolean Algebra: DeMorgans Theorem

Augustus De Morgan (27 June 1806 — 18 March 1871)

15a $$\overline{x \cdot y} = \overline{x} + \overline{y} \qquad (30)$$

15b $$\overline{x + y} = \overline{x} \cdot \overline{y} \qquad (31)$$

16a $x + \overline{x} \cdot y = x + y$

16b $x \cdot (\overline{x} + y) = x \cdot y$

- Huntington's basic postulates-5,8,10,12

$$\begin{aligned} x \cdot 0 &= 0 \\ x + 1 &= 1 \end{aligned}$$

$$\begin{aligned} x \cdot \overline{x} &= 0 \\ x + \cdot \overline{x} &= 1 \end{aligned}$$

$$\begin{aligned} x \cdot y &= y \cdot x \\ x + y &= y + x \end{aligned}$$

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

$$x + y \cdot z = (x + y) \cdot (x + z)$$

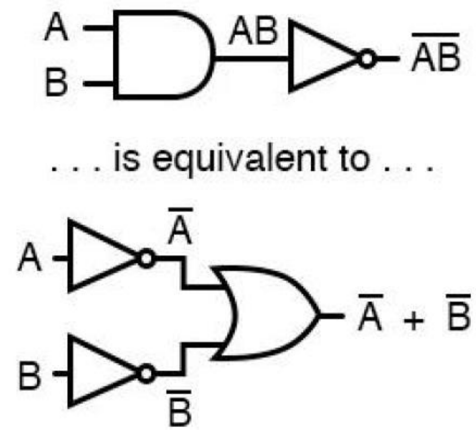
Proof of DeMorgan's Theorem using TruthTable

$$\overline{x \cdot y} = \overline{x} + \overline{y} \qquad (30)$$

$$\overline{x + y} = \overline{x} \cdot \overline{y} \qquad (31)$$

x	y	$x \cdot y$	$\overline{x \cdot y}$	$\overline{x}$	$\overline{y}$	$\overline{x} + \overline{y}$
0	0	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	1	1	0	0	0	0

LHS: $x \cdot y$, $\overline{x \cdot y}$ — RHS: $\overline{x}$, $\overline{y}$, $\overline{x} + \overline{y}$



$$\mathbf{\overline{AB} = \overline{A} + \overline{B}}$$

Proof of DeMorgan's Theorem using TruthTable

$$\overline{x \cdot y} = \overline{x} + \overline{y} \qquad (30)$$

$$\overline{x + y} = \overline{x} \cdot \overline{y} \qquad (31)$$

X	Y	X+Y	$\overline{X+Y}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

X	Y	$\overline{X}$	$\overline{Y}$	$\overline{X}\bullet\overline{Y}$
0	0	1	1	1
0	1	1	0	0
1	0	0	1	0
1	1	0	0	0

Other examples

1. Proof of 13a(Absorption law)

$X + XY = X\cdot 1 + XY = X(1+Y) = X\cdot 1 = X$

- Use 6a 12a 5b

2. Proof of 16a

$X+X'Y = (X+X')(X+Y) = 1\cdot (X+Y) = X+Y$

- Use 12b(distributive) 8b 6a

3. Proof

$XY+XY' = X(Y + Y') = X\cdot 1 = X$

- Use 12a(distributive) 8b 6a

Further Examples

4. $X\cdot (X+Y)=X\cdot X+X\cdot Y= X+XY=X(1+Y)=X\cdot 1=X$
   - Use 12b(distributive) 7a 12a 5b

5. $(X+Y)\cdot(X+Y')=XX+XY'+XY+YY'=$

   $X+XY'+XY+0=X(1+Y'+Y)=X\cdot 1=X$

6. $X(X'+Y) = XX'+XY = 0 + XY = XY$

$$\bar{A}BC + A\bar{B}C + AB\bar{C} + ABC$$

$\downarrow$ Factoring BC out of 1st and 4th terms

$$BC(\bar{A} + A) + A\bar{B}C + AB\bar{C}$$

$\downarrow$ Applying identity $A + \bar{A} = 1$

$$BC(1) + A\bar{B}C + AB\bar{C}$$

$\downarrow$ Applying identity $1A = A$

$$BC + A\bar{B}C + AB\bar{C}$$

$\downarrow$ Factoring B out of 1st and 3rd terms

$$B(C + A\bar{C}) + A\bar{B}C$$

$\downarrow$ Applying rule $A + \bar{A}B = A + B$ to the $C + A\bar{C}$ term

$$B(C + A) + A\bar{B}C$$

$\downarrow$ Distributing terms

$$BC + AB + A\bar{B}C$$

$\downarrow$ Factoring A out of 2nd and 3rd terms

$$BC + A(B + \bar{B}C)$$

$\downarrow$ Applying rule $A + \bar{A}B = A + B$ to the $B + \bar{B}C$ term

$$BC + A(B + C)$$

$\downarrow$ Distributing terms

$$BC + AB + AC$$

or Simplified result

$$AB + BC + AC$$

Proof of Consensus Theorem

- Consensus Theorem gives us the relationship

  $XY + X'Z + YZ = XY + X'Z$

  - Note that in doing the reduction the first step is to and in a 1 to the YZ term. That 1 is in the form, (X+X').
  - Try it.

The consensus theorem can be stated as follows:

$$XY + X'Z + YZ = XY + X'Z$$

Proof

$$XY + X'Z + YZ \rightarrow XY + X'Z + (X + X')YZ$$

$$\rightarrow XY + X'Z + XYZ + X'YZ$$

$$\rightarrow (XY + XYZ) + (X'Z + X'YZ)$$

$$\rightarrow XY(1 + Z) + X'Z(1 + Y)$$

$$\rightarrow XY + X'Z$$

Consensus theorem Proof–POS form

$$(A + B)(B' + C)(A + C) = (A + B)(B' + C)$$

$$\begin{aligned}
\text{LHS} &= (A + B)(B' + C)(A + C) \\
&= (A + B)(B' + C)(A + C + 0) \\
&= (A + B)(B' + C)(A + B.B' + C) \\
&= (A + B)(B' + C)(A + B + C)(A + B' + C) \\
&= [(A + B)(A + B + C)]\,[(B' + C)(A + B' + C)] \\
&= (A + B)\,(B' + C) = \text{RHS}
\end{aligned}$$

Examples

$VX'Y + WXZ + VWYZ =$

$VX'Y + WXZ + VWYZ$

$= VX'Y + WXZ$

$(A + B + E')(E + F + G')(A + B + F + G') =$

$(A + B + E')(E + F + G')(A + B + F + G')$

$= (A + B + E')(E + F + G)$

Simplify

- Simplify F=X'YZ+X'YZ'+XZ

$$F = \overline{X}YZ + \overline{X}Y\overline{Z} + XZ$$

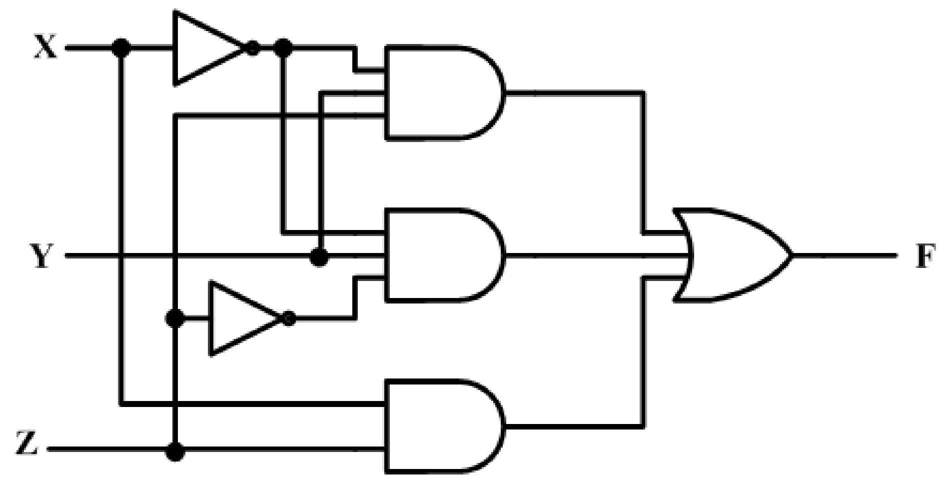
$$= \overline{X}Y(Z + \overline{Z}) + XZ \qquad \text{by identity 14}$$

$$= \overline{X}Y \bullet 1 + XZ \qquad \text{by identity 7}$$

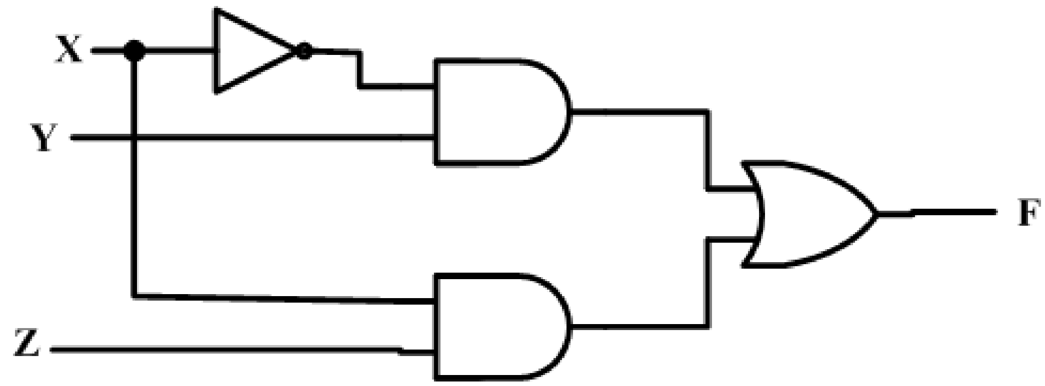
$$= \overline{X}Y + XZ \qquad \text{by identity 2}$$

Affect on implementation

- F = X'YZ + X'YZ' + XZ



- Reduces to F = X'Y + XZ



Let us prove the validity of the logic equation

$$(x_1 + x_3) \cdot (\overline{x}_1 + \overline{x}_3) = x_1 \cdot \overline{x}_3 + \overline{x}_1 \cdot x_3$$

The left-hand side can be manipulated as follows. Using the distributive property, 12*a*, gives

$$\text{LHS} = (x_1 + x_3) \cdot \overline{x}_1 + (x_1 + x_3) \cdot \overline{x}_3$$

Applying the distributive property again yields

$$\text{LHS} = x_1 \cdot \overline{x}_1 + x_3 \cdot \overline{x}_1 + x_1 \cdot \overline{x}_3 + x_3 \cdot \overline{x}_3$$

Note that the distributive property allows ANDing the terms in parenthesis in a way analogous to multiplication in ordinary algebra. Next, according to theorem 8*a*, the terms $x_1 \cdot \overline{x}_1$ and $x_3 \cdot \overline{x}_3$ are both equal to 0. Therefore,

$$\text{LHS} = 0 + x_3 \cdot \overline{x}_1 + x_1 \cdot \overline{x}_3 + 0$$

From 6*b* it follows that

$$\text{LHS} = x_3 \cdot \overline{x}_1 + x_1 \cdot \overline{x}_3$$

Finally, using the commutative property, 10*a* and 10*b*, this becomes

$$\text{LHS} = x_1 \cdot \overline{x}_3 + \overline{x}_1 \cdot x_3$$

which is the same as the right-hand side of the initial equation.

- Try

$$x_1 \cdot \overline{x}_3 + \overline{x}_2 \cdot \overline{x}_3 + x_1 \cdot x_3 + \overline{x}_2 \cdot x_3 = \overline{x}_1 \cdot \overline{x}_2 + x_1 \cdot x_2 + x_1 \cdot \overline{x}_2$$

- Solution:

$$x_1 \cdot \overline{x}_3 + \overline{x}_2 \cdot \overline{x}_3 + x_1 \cdot x_3 + \overline{x}_2 \cdot x_3 = \overline{x}_1 \cdot \overline{x}_2 + x_1 \cdot x_2 + x_1 \cdot \overline{x}_2$$

The left-hand side can be manipulated as follows

$$\begin{aligned} \text{LHS} &= x_1 \cdot \overline{x}_3 + x_1 \cdot x_3 + \overline{x}_2 \cdot \overline{x}_3 + \overline{x}_2 \cdot x_3 && \text{using } 10b \\ &= x_1 \cdot (\overline{x}_3 + x_3) + \overline{x}_2 \cdot (\overline{x}_3 + x_3) && \text{using } 12a \\ &= x_1 \cdot 1 + \overline{x}_2 \cdot 1 && \text{using } 8b \\ &= x_1 + \overline{x}_2 && \text{using } 6a \end{aligned}$$

The right-hand side can be manipulated as

$$\begin{aligned} \text{RHS} &= \overline{x}_1 \cdot \overline{x}_2 + x_1 \cdot (x_2 + \overline{x}_2) && \text{using } 12a \\ &= \overline{x}_1 \cdot \overline{x}_2 + x_1 \cdot 1 && \text{using } 8b \\ &= \overline{x}_1 \cdot \overline{x}_2 + x_1 && \text{using } 6a \\ &= x_1 + \overline{x}_1 \cdot \overline{x}_2 && \text{using } 10b \\ &= x_1 + \overline{x}_2 && \text{using } 16a \end{aligned}$$

Complement of a function

- In real implementation sometimes the complement of a function is needed.
  - Have F=X'YZ'+X'Y'Z

$$F = \overline{X}Y\overline{Z} + \overline{X}\,\overline{Y}Z$$

$$\overline{F} = \overline{\overline{X}Y\overline{Z} + \overline{X}\,\overline{Y}Z}$$

$$= (\overline{\overline{X}Y\overline{Z}}) \bullet (\overline{\overline{X}\,\overline{Y}Z})$$

$$= (X+\overline{Y}+Z) \bullet (X+Y+\overline{Z})$$

How Duals and Complement are related

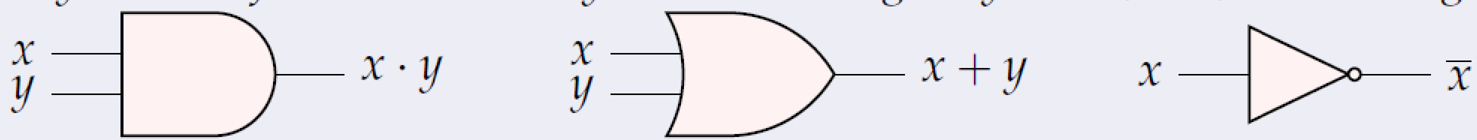
- What is meant by the dual of a function?
  - The *dual* of a function is obtained by interchanging OR and AND operations and replacing 1s and 0s with 0s and 1s.
- Shortcut to getting function complement
  - Starting with the equation on the previous slide
  - Generate the dual $F=(X'+Y+Z')(X'+Y'+Z)$
  - Complement each literal to get:
  - $F'=(X+Y'+Z)(X+Y+Z')$

LECTURE 4

Synthesis of simple circuits

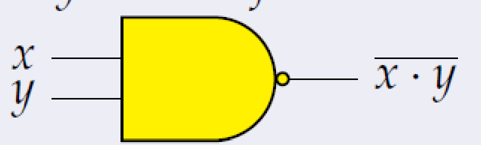
Theorem

Any Boolean function can be synthesized using only AND, OR, and NOT gates.



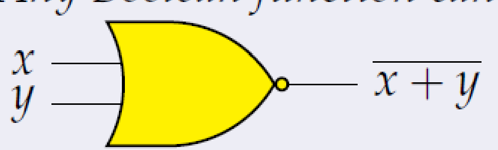
Theorem

Any Boolean function can be synthesized using only NAND gates.



Theorem

Any Boolean function can be synthesized using only NOR gates.



Synthesis using AND,OR, NOT Gates

Given a Boolean function f given in the form of a truth table, the expression that realizes f can be obtained:

- (SUM-OF-PRODUCTS) by considering rows for which $f = 1$, or
- (PRODUCTS-OF-SUMS) by considering rows for which $f = 0$.

Sum of Products Canonical form

- For a function of n variables a minterm is a product term in which each of the variable occur exactly once. For example: $x_0 \cdot x_1 \cdot \overline{x_2}$ and $\overline{x_0} \cdot \overline{x_1} \cdot x_2$.
- A function f can be represented by an expression that is sum of minterms.
- A logical expression that consists of product terms that are summed (ORed) together is called a sum-of-product form.
- If each of the product term is a minterm, then the expression is called canonical sum-of-product expression.
- Notice that two logically equivalent functions will have the same canonical representation.

Two Variable Minterms

Minterms			
x	y	term	notation
0	0	x'.y'	m0
0	1	x'.y	m1
1	0	x.y'	m2
1	1	x.y	m3

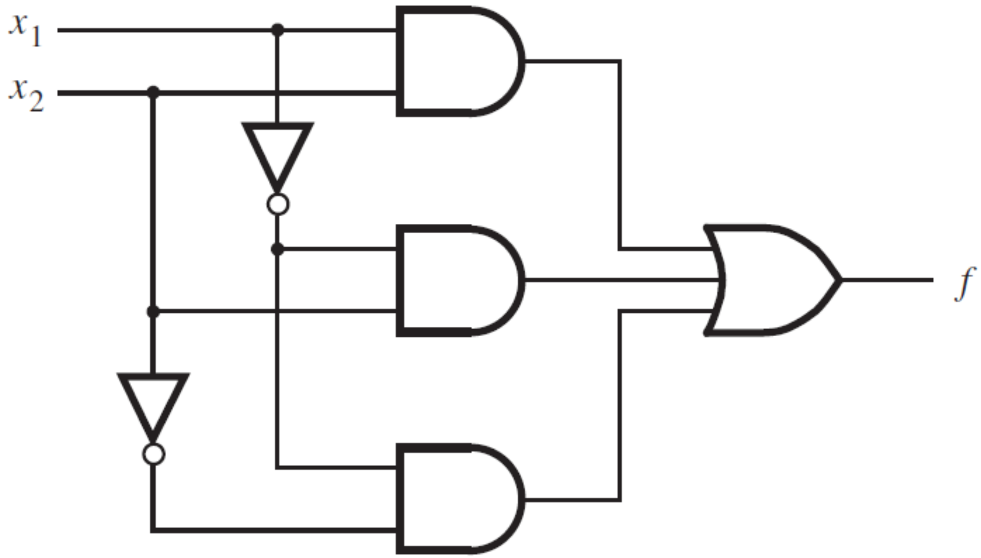
Simple problem of synthesis

- Assume that $x1$ and $x2$ represent the states of two switches, either of which may produce a 0 or 1.
- The function of the circuit is to continuously monitor the state of the switches and to produce an output logic value 1 whenever the switches $(x1, x2)$ are in states $(0, 0)$, $(0, 1)$, or $(1, 1)$. If the state of the switches is $(1, 0)$, the output should be 0.
- We can express the required behavior using a truth table,

x_1	x_2	$f(x_1, x_2)$	
0	0	1	$\leftarrow \bar{x}_1\bar{x}_2$
0	1	1	$\leftarrow \bar{x}_1 x_2$
1	0	0	
1	1	1	$\leftarrow x_1 x_2$

$$f(x_1, x_2) = x_1x_2 + \bar{x}_1\bar{x}_2 + \bar{x}_1x_2$$

Canonical SOP –implementation using Logic Gates



(a) Canonical sum-of-products

$$f(x_1, x_2) = x_1x_2 + \bar{x}_1\bar{x}_2 + \bar{x}_1x_2$$

The process whereby we begin with a description of the desired functional behavior and then generate a circuit that realizes this behavior is called *synthesis*.

Canonical SOP expression can be reduced using Boolean Algebra

$$f(x_1, x_2) = x_1x_2 + \overline{x}_1\overline{x}_2 + \overline{x}_1x_2$$

1. Using 7b,third Product term is replicated

$$f(x_1, x_2) = x_1x_2 + \overline{x}_1\overline{x}_2 + \overline{x}_1x_2 + \overline{x}_1x_2$$

2. Using the commutative property 10*b* to interchange the second and third product terms gives

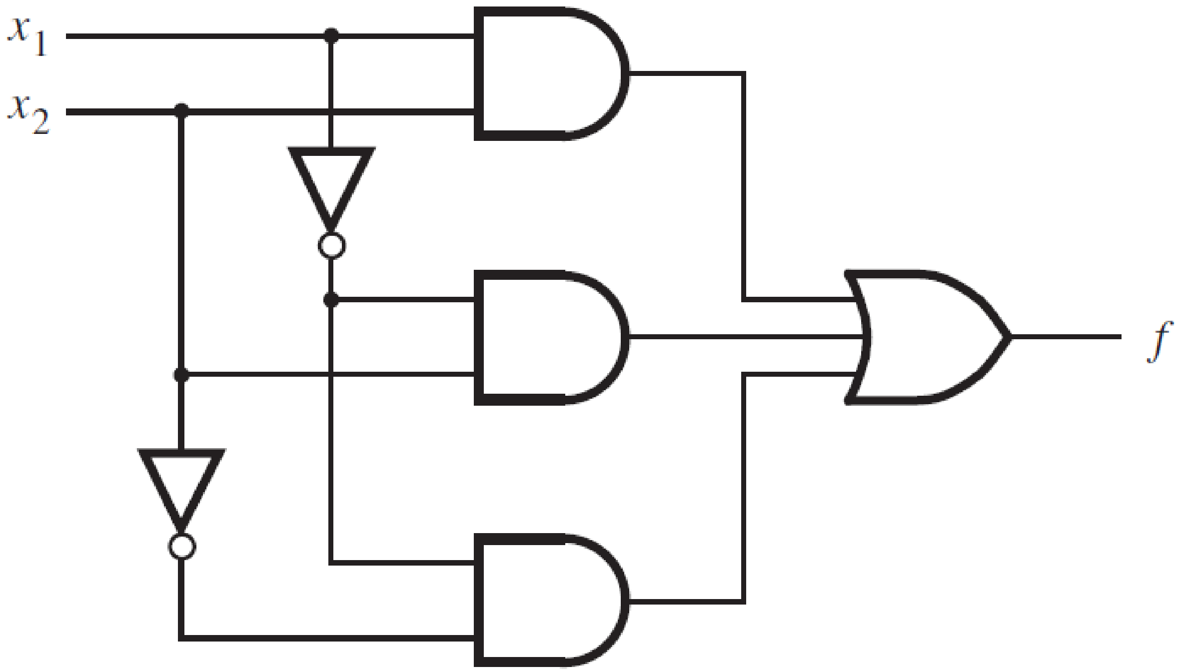
$$f(x_1, x_2) = x_1x_2 + \overline{x}_1x_2 + \overline{x}_1\overline{x}_2 + \overline{x}_1x_2$$

3.The distributive property 12*a*

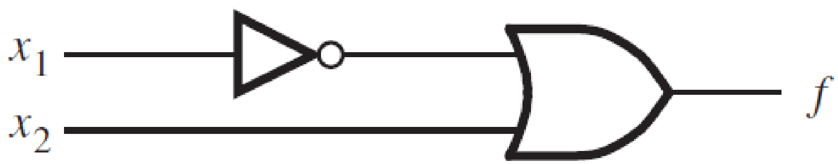
$$f(x_1, x_2) = (x_1 + \overline{x}_1)x_2 + \overline{x}_1(\overline{x}_2 + x_2)$$

4.Using 8b and 6a

$$f(x_1, x_2) = 1 \cdot x_2 + \overline{x}_1 \cdot 1 \Rightarrow f(x_1, x_2) = x_2 + \overline{x}_1$$



(a) Canonical sum-of-products



(b) Minimal-cost realization

Three Variable Minterms

Row number	x_1	x_2	x_3	Minterm
0	0	0	0	$m_0 = \overline{x}_1\overline{x}_2\overline{x}_3$
1	0	0	1	$m_1 = \overline{x}_1\overline{x}_2x_3$
2	0	1	0	$m_2 = \overline{x}_1x_2\overline{x}_3$
3	0	1	1	$m_3 = \overline{x}_1x_2x_3$
4	1	0	0	$m_4 = x_1\overline{x}_2\overline{x}_3$
5	1	0	1	$m_5 = x_1\overline{x}_2x_3$
6	1	1	0	$m_6 = x_1x_2\overline{x}_3$
7	1	1	1	$m_7 = x_1x_2x_3$

- Consider three variable function and synthesize the circuit for it.

Row number	x_1	x_2	x_3	$f(x_1, x_2, x_3)$
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

- Consider three variable function and synthesize the circuit for it.

Row number	x_1	x_2	x_3	$f(x_1, x_2, x_3)$
0	0	0	0	0
1	0	0	1	1
2	0	1	0	0
3	0	1	1	0
4	1	0	0	1
5	1	0	1	1
6	1	1	0	1
7	1	1	1	0

$$f(x_1, x_2, x_3) = \sum (m_1, m_4, m_5, m_6)$$

$$f(x_1, x_2, x_3) = \sum m(1, 4, 5, 6)$$

- Consider three variable function and synthesize the circuit for it.

Row number	x_1	x_2	x_3	$f(x_1, x_2, x_3)$	
0	0	0	0	0	
1	0	0	1	1	$\overline{x}_1\overline{x}_2x_3$
2	0	1	0	0	
3	0	1	1	0	
4	1	0	0	1	$x_1\overline{x}_2\overline{x}_3$
5	1	0	1	1	$x_1\overline{x}_2x_3$
6	1	1	0	1	$x_1x_2\overline{x}_3$
7	1	1	1	0	

Canonical SOP expression

$$f(x_1, x_2, x_3) = \overline{x}_1\overline{x}_2x_3 + x_1\overline{x}_2\overline{x}_3 + x_1\overline{x}_2x_3 + x_1x_2\overline{x}_3$$

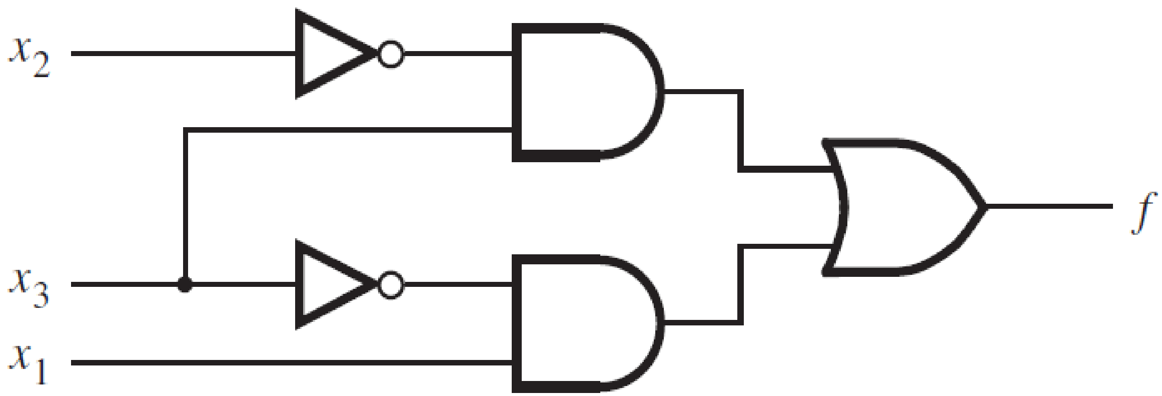
- Draw circuit diagram
- Reduce it and again draw the reduced circuit diagram.
- Compare the cost-A good indication of the *cost* of a logic circuit is the total number of gates plus the total number of inputs to all gates in the circuit.

$$f(x_1, x_2, x_3) = \bar{x}_1\bar{x}_2x_3 + x_1\bar{x}_2\bar{x}_3 + x_1\bar{x}_2x_3 + x_1x_2\bar{x}_3$$

$$\begin{aligned} f &= (\bar{x}_1 + x_1)\bar{x}_2x_3 + x_1(\bar{x}_2 + x_2)\bar{x}_3 \\ &= 1 \cdot \bar{x}_2x_3 + x_1 \cdot 1 \cdot \bar{x}_3 \\ &= \bar{x}_2x_3 + x_1\bar{x}_3 \end{aligned}$$

$$= \bar{x}_2x_3 + x_1\bar{x}_3$$

Cost:13(5 Gates & 8 inputs) vs 27 for canonical exp ckt realization



(a) A minimal sum-of-products realization

Example

Another example:

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$Sum = \overline{A}\,\overline{B}\,Cin + \overline{A}\,B\,\overline{Cin} + A\,\overline{B}\,\overline{Cin} + A\,B\,Cin$$

$$Cout = \overline{A}\,B\,Cin + A\,\overline{B}\,Cin + A\,B\,\overline{Cin} + A\,B\,Cin$$

Product of Sum Canonical Form

- For a function of n variables a maxterm is a sum term in which each of the variable occur exactly once. For example

$$(x_0 + x_1 + \overline{x_2}) \text{ and } (\overline{x_0} + \overline{x_1} + x_2).$$

- A function f can be represented by an expression that is product of maxterms.
- A logical expression that consists of sum terms that are factors of logical product (AND) is called a product-of-sum form.
- If each of the sum term is a maxterm, then the expression is called canonical product-of-sum expression.
- Notice that two logically equivalent functions will have the same canonical product-of-sum representation.

Maxterms

Row number	x_1	x_2	x_3	Minterm	Maxterm
0	0	0	0	$m_0 = \bar{x}_1\bar{x}_2\bar{x}_3$	$M_0 = x_1 + x_2 + x_3$
1	0	0	1	$m_1 = \bar{x}_1\bar{x}_2x_3$	$M_1 = x_1 + x_2 + \bar{x}_3$
2	0	1	0	$m_2 = \bar{x}_1x_2\bar{x}_3$	$M_2 = x_1 + \bar{x}_2 + x_3$
3	0	1	1	$m_3 = \bar{x}_1x_2x_3$	$M_3 = x_1 + \bar{x}_2 + \bar{x}_3$
4	1	0	0	$m_4 = x_1\bar{x}_2\bar{x}_3$	$M_4 = \bar{x}_1 + x_2 + x_3$
5	1	0	1	$m_5 = x_1\bar{x}_2x_3$	$M_5 = \bar{x}_1 + x_2 + \bar{x}_3$
6	1	1	0	$m_6 = x_1x_2\bar{x}_3$	$M_6 = \bar{x}_1 + \bar{x}_2 + x_3$
7	1	1	1	$m_7 = x_1x_2x_3$	$M_7 = \bar{x}_1 + \bar{x}_2 + \bar{x}_3$

Maxterms

- The principle of duality suggests that if it is possible to synthesize a function f by considering the rows in the truth table for which $f = 1$, then it should also be possible to synthesize f by considering the rows for which $f = 0$.
- This alternative approach uses the complements of minterms, which are called *maxterms*.

- Consider three variable function and synthesize the circuit for it using Maxterms

Row number	x_1	x_2	x_3	$f(x_1, x_2, x_3)$	
0	0	0	0	0	$(x_1 + x_2 + x_3)$
1	0	0	1	1	
2	0	1	0	0	$(x_1 + \overline{x}_2 + x_3)$
3	0	1	1	0	$(x_1 + \overline{x}_2 + \overline{x}_3)$
4	1	0	0	1	
5	1	0	1	1	
6	1	1	0	1	
7	1	1	1	0	$(\overline{x}_1 + \overline{x}_2 + \overline{x}_3)$

$$f(x_1, x_2, x_3) = \Pi(M_0, M_2, M_3, M_7)$$

$$f(x_1, x_2, x_3) = \Pi M(0, 2, 3, 7)$$

- Consider three variable function and synthesize the circuit for it using Maxterms

Row number	x_1	x_2	x_3	$f(x_1, x_2, x_3)$	
0	0	0	0	0	$(x_1 + x_2 + x_3)$
1	0	0	1	1	
2	0	1	0	0	$(x_1 + \overline{x}_2 + x_3)$
3	0	1	1	0	$(x_1 + \overline{x}_2 + \overline{x}_3)$
4	1	0	0	1	
5	1	0	1	1	
6	1	1	0	1	
7	1	1	1	0	$(\overline{x}_1 + \overline{x}_2 + \overline{x}_3)$

Canonical POS expression

$$f(x_1, x_2, x_3) = M_0 \cdot M_2 \cdot M_3 \cdot M_7$$

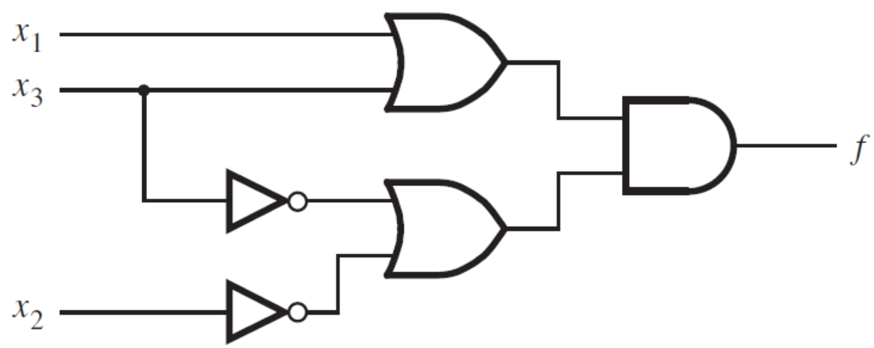
$$= (x_1 + x_2 + x_3)(x_1 + \overline{x}_2 + x_3)(x_1 + \overline{x}_2 + \overline{x}_3)(\overline{x}_1 + \overline{x}_2 + \overline{x}_3)$$

Reduce it and draw circuit diagram

$$f(x_1, x_2, x_3) = (x_1 + x_2 + x_3)(x_1 + \overline{x}_2 + x_3)(x_1 + \overline{x}_2 + \overline{x}_3)(\overline{x}_1 + \overline{x}_2 + \overline{x}_3)$$

$$f = ((x_1 + x_3) + x_2)((x_1 + x_3) + \overline{x}_2)(x_1 + (\overline{x}_2 + \overline{x}_3))(\overline{x}_1 + (\overline{x}_2 + \overline{x}_3))$$

$$f = (x_1 + x_3)(\overline{x}_2 + \overline{x}_3)$$ Using 14b-combining property



(b) A minimal product-of-sums realization

Consider the function

$$f(x_1, x_2, x_3) = \sum m(2, 3, 4, 6, 7)$$

Q1>Find the cost of Canonical SOP expression.
Q2>Reduce this expression and find cost again.
Q3>Write corresponding Maxterm list-canonical POS form
Q4>Reduce canonical POS expression and find the cost.
Q5>Compare Q2 and Q4 costs

$$
\begin{aligned}
f &= m_2 + m_3 + m_4 + m_6 + m_7 \\
&= \overline{x}_1 x_2 \overline{x}_3 + \overline{x}_1 x_2 x_3 + x_1 \overline{x}_2 \overline{x}_3 + x_1 x_2 \overline{x}_3 + x_1 x_2 x_3
\end{aligned}
$$

$$
\begin{aligned}
f &= \overline{x}_1 x_2 (\overline{x}_3 + x_3) + x_1 (\overline{x}_2 + x_2) \overline{x}_3 + x_1 x_2 (\overline{x}_3 + x_3) \\
&= \overline{x}_1 x_2 + x_1 \overline{x}_3 + x_1 x_2 \\
&= (\overline{x}_1 + x_1) x_2 + x_1 \overline{x}_3 \\
&= x_2 + x_1 \overline{x}_3
\end{aligned}
$$

$$f(x_1, x_2, x_3) = \Pi M(0, 1, 5)$$

Then, the canonical POS expression is derived as

$$
\begin{aligned}
f &= M_0 \cdot M_1 \cdot M_5 \\
&= (x_1 + x_2 + x_3)(x_1 + x_2 + \overline{x}_3)(\overline{x}_1 + x_2 + \overline{x}_3)
\end{aligned}
$$

A simplified POS expression can be derived as

$$
\begin{aligned}
f &= (x_1 + x_2 + x_3)(x_1 + x_2 + \overline{x}_3)(x_1 + x_2 + \overline{x}_3)(\overline{x}_1 + x_2 + \overline{x}_3) \\
&= ((x_1 + x_2) + x_3)((x_1 + x_2) + \overline{x}_3)(x_1 + (x_2 + \overline{x}_3))(\overline{x}_1 + (x_2 + \overline{x}_3)) \\
&= ((x_1 + x_2) + x_3 \overline{x}_3)(x_1 \overline{x}_1 + (x_2 + \overline{x}_3)) \\
&= (x_1 + x_2)(x_2 + \overline{x}_3)
\end{aligned}
$$

Find minimum cost SOP expression for this function

Suppose that a four-variable function is defined by

$$f(x_1, x_2, x_3, x_4) = \sum m(3, 7, 9, 12, 13, 14, 15)$$

The canonical SOP expression for this function is

$$f = \overline{x}_1\overline{x}_2x_3x_4 + \overline{x}_1x_2x_3x_4 + x_1\overline{x}_2\overline{x}_3x_4 + x_1x_2\overline{x}_3\overline{x}_4 + x_1x_2\overline{x}_3x_4 + x_1x_2x_3\overline{x}_4 + x_1x_2x_3x_4$$

A simpler SOP expression can be obtained as follows

$$\begin{aligned} f &= \overline{x}_1(\overline{x}_2 + x_2)x_3x_4 + x_1(\overline{x}_2 + x_2)\overline{x}_3x_4 + x_1x_2\overline{x}_3(\overline{x}_4 + x_4) + x_1x_2x_3(\overline{x}_4 + x_4) \\ &= \overline{x}_1x_3x_4 + x_1\overline{x}_3x_4 + x_1x_2\overline{x}_3 + x_1x_2x_3 \\ &= \overline{x}_1x_3x_4 + x_1\overline{x}_3x_4 + x_1x_2(\overline{x}_3 + x_3) \\ &= \overline{x}_1x_3x_4 + x_1\overline{x}_3x_4 + x_1x_2 \end{aligned}$$

Problem: Design the minimum-cost product-of-sums expression for the function $f(x_1, x_2, x_3, x_4) = \sum m(0, 2, 4, 5, 6, 7, 8, 10, 12, 14, 15)$.

Solution: The function is defined in terms of its minterms. To find a POS expression we should start with the definition in terms of maxterms, which is $f = \Pi M(1, 3, 9, 11, 13)$. Thus,

$$\begin{aligned} f &= M_1 \cdot M_3 \cdot M_9 \cdot M_{11} \cdot M_{13} \\ &= (x_1 + x_2 + x_3 + \overline{x}_4)(x_1 + x_2 + \overline{x}_3 + \overline{x}_4)(\overline{x}_1 + x_2 + x_3 + \overline{x}_4)(\overline{x}_1 + x_2 + \overline{x}_3 + \overline{x}_4)(\overline{x}_1 + \overline{x}_2 + x_3 + \overline{x}_4) \end{aligned}$$

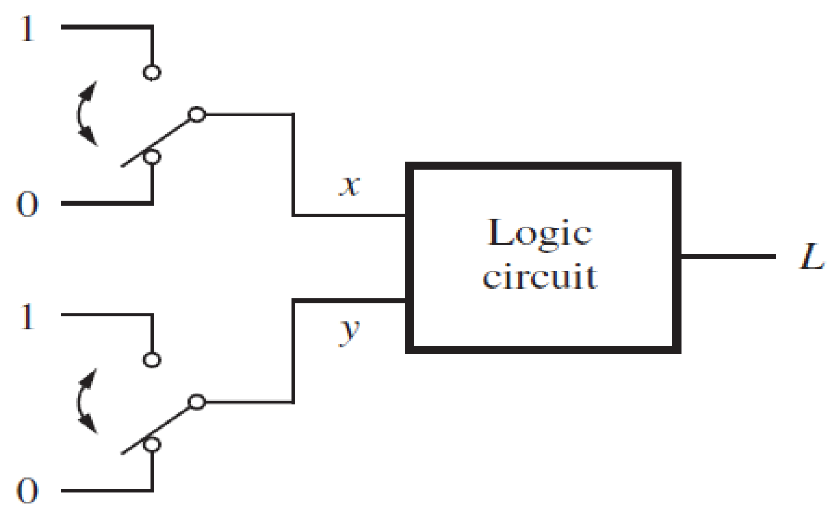
We can rewrite the product of the first two maxterms as

$$\begin{aligned} M_1 \cdot M_3 &= (x_1 + x_2 + \overline{x}_4 + x_3)(x_1 + x_2 + \overline{x}_4 + \overline{x}_3) && \text{using commutative property } 10b \\ &= x_1 + x_2 + \overline{x}_4 + x_3\overline{x}_3 && \text{using distributive property } 12b \\ &= x_1 + x_2 + \overline{x}_4 + 0 && \text{using theorem } 8a \\ &= x_1 + x_2 + \overline{x}_4 && \text{using theorem } 6b \end{aligned}$$

Similarly, $M_9 \cdot M_{11} = \overline{x}_1 + x_2 + \overline{x}_4$. Now, we can use M_9 again, according to property $7a$, to derive $M_9 \cdot M_{13} = \overline{x}_1 + x_3 + \overline{x}_4$. Hence

$$f = (x_1 + x_2 + \overline{x}_4)(\overline{x}_1 + x_2 + \overline{x}_4)(\overline{x}_1 + x_3 + \overline{x}_4)$$

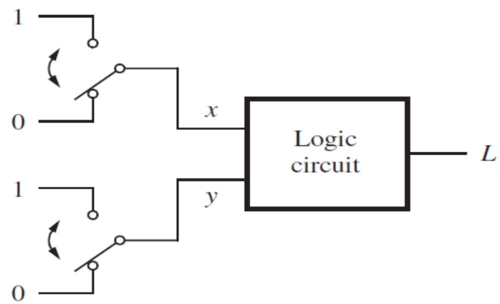
Try this problem: Write Truth Table and draw circuit diagram



(a) Two switches that control a light

The required behavior is that the light should be on only if one, but not both, of the toggle switches is in the top position.

Truth Table and Circuit diagram

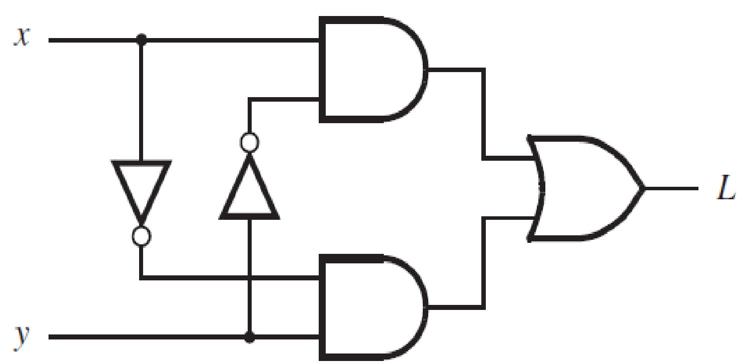


(a) Two switches that control a light

$L = \overline{x} \cdot y + x \cdot \overline{y}$, we can write $L = x \oplus y$.

x	y	L
0	0	0
0	1	1
1	0	1
1	1	0

(b) Truth table



(c) Logic network

(d) XOR gate symbol

Synthesis Example

A circuit that controls a given digital system has three inputs: $x1$, $x2$, and $x3$.It has to recognize three different conditions:

- Condition A is true if $x3$ is true and either $x1$ is true or $x2$ is false
- Condition B is true if $x1$ is true and either $x2$ or $x3$ is false
- Condition C is true if $x2$ is true and either $x1$ is true or $x3$ is false

The control circuit must produce an output of 1 if at least two of the conditions A, B, and C are true. Design the simplest circuit that can be used for this purpose.

Solution: Using 1 for true and 0 for false, we can express the three conditions as follows:

$$\begin{aligned} A = x_3(x_1 + \overline{x}_2) &= x_3x_1 + x_3\overline{x}_2 \\ B = x_1(\overline{x}_2 + \overline{x}_3) &= x_1\overline{x}_2 + x_1\overline{x}_3 \\ C = x_2(x_1 + \overline{x}_3) &= x_2x_1 + x_2\overline{x}_3 \end{aligned}$$

Then, the desired output of the circuit can be expressed as $f = AB + AC + BC$. These product terms can be determined as:

$$\begin{aligned} AB &= (x_3x_1 + x_3\overline{x}_2)(x_1\overline{x}_2 + x_1\overline{x}_3) \\ &= x_3x_1x_1\overline{x}_2 + x_3x_1x_1\overline{x}_3 + x_3\overline{x}_2x_1\overline{x}_2 + x_3\overline{x}_2x_1\overline{x}_3 \\ &= x_3x_1\overline{x}_2 + 0 + x_3\overline{x}_2x_1 + 0 \\ &= x_1\overline{x}_2x_3 \end{aligned}$$

$$\begin{aligned} AC &= (x_3x_1 + x_3\overline{x}_2)(x_2x_1 + x_2\overline{x}_3) \\ &= x_3x_1x_2x_1 + x_3x_1x_2\overline{x}_3 + x_3\overline{x}_2x_2x_1 + x_3\overline{x}_2x_2\overline{x}_3 \\ &= x_3x_1x_2 + 0 + 0 + 0 \\ &= x_1x_2x_3 \end{aligned}$$

$$\begin{aligned} BC &= (x_1\overline{x}_2 + x_1\overline{x}_3)(x_2x_1 + x_2\overline{x}_3) \\ &= x_1\overline{x}_2x_2x_1 + x_1\overline{x}_2x_2\overline{x}_3 + x_1\overline{x}_3x_2x_1 + x_1\overline{x}_3x_2\overline{x}_3 \\ &= 0 + 0 + x_1\overline{x}_3x_2 + x_1\overline{x}_3x_2 \\ &= x_1x_2\overline{x}_3 \end{aligned}$$

Therefore, f can be written as

$$\begin{aligned} f &= x_1\overline{x}_2x_3 + x_1x_2x_3 + x_1x_2\overline{x}_3 \\ &= x_1(\overline{x}_2 + x_2)x_3 + x_1x_2(x_3 + \overline{x}_3) \\ &= x_1x_3 + x_1x_2 \\ &= x_1(x_3 + x_2) \end{aligned}$$